

Hitesh Kolluru

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with nearly 4 years of production experience and an MS in Software Engineering, building backends, APIs, and client-facing applications in **Java, Spring Boot, React, and Python**. Shipped C#/.NET iOS and Android apps for a major healthcare provider that reached **100K+ Google Play downloads** and a **4.7-star App Store rating** across 12K ratings. Recent work spans a Spring Boot GraphQL platform architected and deployed solo on **AWS** with keyless CI/CD, a service owned end to end inside an **18-microservice platform**, and a research knowledge-graph system that answers plain-English questions over linked datasets **14x more accurately** than the strongest of seven baseline LLMs.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, C#, SQL

Frontend: React, Next.js, Vue, Vite, Apollo Client, D3, Recharts, HTML/CSS

Backend & APIs: Spring Boot, Node.js (Nest.js), FastAPI, Flask, REST, GraphQL, Microservices, Distributed Systems, MVVM

Cloud & DevOps: AWS (EC2, RDS, S3, CloudFront, CloudWatch, IAM), Docker, GitHub Actions, Azure DevOps, CI/CD, GitHub OIDC

Databases: PostgreSQL, MongoDB, MySQL, ChromaDB, Ontotext GraphDB, SPARQL

AI & ML: RAG, Hybrid Retrieval (BM25 + Semantic), Embeddings, LLM Evaluation, MCP, Agentic Systems, PyTorch, Transformers, NLTK, spaCy, OpenAI / Gemini / Groq / Ollama, LiteLLM

Practices: Agile/Scrum, Kanban, Unit Testing, TDD, Code Review, JWT, RBAC, Design Patterns, Data Structures & Algorithms

Tools: Git, Jira, Taiga, Gradle, Maven, Postman

WORK EXPERIENCE

Capgemini — Software Development Engineer

Apr 2021 – Jun 2024

Clients: Banner Health (healthcare), internal inventory platform

- Shipped **native iOS and Android UI** for a major healthcare provider's patient app in **C#/.NET** (Xamarin) on an **MVVM** architecture, reaching **100K+ Google Play downloads** and a **4.7-star App Store rating** across 12K ratings.
- Engineered a **server-driven UI** rendering screens at runtime from API response schemas, letting the team ship new questionnaires and forms **without an app release or store review cycle**.
- Delivered an internal clinician directory and messaging application, giving doctors searchable access to colleagues across specialty, practice location, and schedule.
- Led R&D and technical design for a Xamarin to **.NET MAUI** migration, driving it to **roughly 75% completion** and producing migration documentation **adopted by other teams**.
- Integrated **10+ backend-owned REST APIs** under fixed contracts and reviewed peer pull requests across a **20+ engineer organization**, shipping through Azure DevOps CI/CD and Jira.

Arizona State University — Graduate Course Grader (SER515)

Aug 2025 – Dec 2025

Foundations of Software Engineering

Tempe, AZ

- Cut grading turnaround from **up to a week to same-day** for **roughly 100 graduate students** by automating the assessment pipeline in **Python** and pandas against the Canvas LMS REST API.
- Assessed Scrum practice, design artifacts, testing, and **PMD static analysis** against the course rubric, personally issuing **roughly 90% of all grades** in the course that semester.
- Wrote per-submission technical feedback naming the specific engineering practice missed, so students could correct process rather than just absorb a deduction.

OTO Qual — Machine Learning Engineer

May 2019 – Jan 2020

Legal technology — startup ceased operations (COVID-19)

- Built the company's **NLP pipeline** as the **sole engineer** — Scrapy crawlers across **100+ sites**, an **NLTK/spaCy** classifier, model training, and a **Flask REST API** — delivering the working product prototype end to end.
- Extended the core **ASP.NET MVC** product with a PDF watermarking utility that stamped processing state onto each document, making handled and unhandled files distinguishable at a glance.

PROJECTS

Booktown API Platform — Full-Stack GraphQL Bookstore | Jun 2026 | Java 17, Spring Boot 3, GraphQL, React, PostgreSQL, AWS

https://github.com/HiteshKolluru/BooktownAPI_Platform | <https://d2jp9hcegs2w8v.cloudfront.net>

- Built and deployed a full-stack GraphQL bookstore **solo** — React storefront (Vite, Apollo Client) over a **Java 17 / Spring Boot 3** API — running **live in production** on AWS today.
- Architected the AWS deployment across EC2, RDS and S3 with CloudFront, serving the frontend from CloudFront edge at **roughly 150ms time-to-first-byte**, with CloudWatch monitoring and H2 for local development.
- Hardened the public API with **JWT** authentication, role-based authorization and a per-IP rate-limiting filter, closing every unauthenticated path before launch.
- Automated **CI/CD in GitHub Actions** with Gradle build, tests and JaCoCo coverage on every push, deploying through **GitHub OIDC** to eliminate long-lived AWS credentials from the pipeline.

GEMR-KG — Global Emerging Markets Risk Knowledge Graph | Oct 2025 – May 2026 | Presented, IADIS IS 2026 (Zagreb)

React, Python, FastAPI, Ontotext GraphDB, SPARQL, OWL/RDF, Docker, LiteLLM

<https://github.com/AGR19/GEMR-KG-Global-Emerging-Market-Risk-Knowledge-Graph>

- Led technical implementation of a 4-person research project, unifying **three siloed World Bank datasets** into a single **OWL ontology** on a SPARQL endpoint and presenting the result at **IADIS IS 2026**.
- Engineered a **Python/FastAPI** service letting non-technical users query the knowledge graph in plain English, answering in **2.5s average** — **fastest of eight systems benchmarked** — with a self-healing retry layer recovering **10.9%** of failed queries.
- Designed the **73-question benchmark** validating the system against seven baseline LLMs, proving **78.1% answer accuracy versus 5.5%** for the strongest baseline — a **14x improvement**.
- Refactored the LLM layer onto **LiteLLM**, making the system provider-agnostic across OpenAI, Anthropic and Gemini and removing single-vendor lock-in.
- Built the **React 19** frontend rendering emerging-market risk on an interactive 3D globe, with Recharts time-series putting CPI and default-rate trends for **six countries** on one screen.
- Containerized the GraphDB instance with **Docker** and authored the SPARQL query templates that became the ground truth for the evaluation benchmark.

AI Copilot for Project Management | ASU Capstone (SER517), Spring 2026 | Client: Aligned Rewards

Python, FastAPI, MCP, OpenAI, Nest.js, MongoDB, Docker

- Extended a client’s project-management chatbot into an agentic assistant that performs real CRUD operations conversationally, collapsing multi-step UI navigation into one request and **delivered to the client** at engagement close.
- Built the **MCP** integration as a tool registry of JSON-schema’d platform operations plus an execution layer, so the model invokes **real actions through structured tool calls** instead of generating free text.
- Implemented teammate name resolution, so a command naming another person resolves to that person’s actual account rather than a best guess.
- Integrated with the client’s multi-tenant, **JWT**-secured Nest.js platform, propagating authentication so the conversational path **could never exceed a user’s role permissions**.
- Maintained conversational state across turns and contributed a voice-command input path alongside the text interface.

Agile Metrics Microservices Platform | ASU SER516, Spring 2025 | Java, Spring Boot, JavaScript, Vue, Docker, REST

- Owned the **Defect Density service** end to end inside an **18-microservice** platform — Java/Spring Boot API, unit tests, visualization and gateway integration — computing code-quality metrics live from Git history.
- **Selected as 1 of 5 engineers** from a **22-person organization** to form the orchestration team, integrating 18 independently built services across Flask, FastAPI, Spring Boot and NestJS under **standardized API contracts**.
- Implemented client-side history caching in vanilla JavaScript, serving repeat repository views **straight from browser cache** with no backend round-trip.
- Containerized services with **Docker** and delivered through Scrum and Kanban in Taiga across timeboxed rounds, working a Git branching and pull-request review workflow.

Resonating Strike — Multi-Collection RAG Pipeline | *Mar – Apr 2026* | *Python, ChromaDB, BM25, Groq, Ollama*

<https://github.com/HiteshKolluru/ResonatingStrike>

- Built a **RAG** pipeline answering analytical questions by retrieving across **four ChromaDB collections** — community reactions, player profiles, champion statistics and tournament results — and synthesizing one grounded answer.
- Implemented **hybrid retrieval** blending BM25 keyword matching with semantic vector search over **1,565 scraped comments**, normalizing both ranking signals into a single relevance score.
- Assembled the knowledge base from Reddit and Riot's Data Dragon API covering **40 players, 8 coaches and 61 champions**, with a cleaning pass stripping bot posts before indexing.
- Served generation through both Groq (hosted) and Ollama (local) backends and wrote an evaluation harness to compare retrieval configurations head to head.

GPT From Scratch — Character-Level Transformer | *Mar – Apr 2026* | *Python, PyTorch, FastAPI, React, D3*

- Implemented a character-level GPT transformer from first principles in **PyTorch** — tokenizer, positional encoding, **multi-head self-attention** and a **4-layer stack with 512-token context** — **without high-level modeling libraries**, to work from the mechanics up rather than from an API down.
- Wrote the training loop and generation pipeline, running **10,000 training iterations** per checkpoint across **three corpora** (Shakespeare, Sherlock Holmes, Lovecraft) to show how training data alone reshapes model output.
- Served the trained model through a **FastAPI** endpoint with **token-level streaming**, backed by a React and D3 frontend for interactive generation.

Music Lyric Analysis — Data Science Pipeline | *ASU SER594, Fall 2024* | *Python, scikit-learn, Word2Vec, pandas*
https://github.com/HiteshKolluru/ser594_23fc_project

- Built an end-to-end data science pipeline testing whether lyrics predict commercial and critical performance, reconciling **three sources**: Billboard Hot 100 charts, Pitchfork review scores and scraped lyrics.
- Engineered features from raw lyrics using **Word2Vec** embeddings and trained KMeans clustering and linear regression models against chart and sentiment outcomes.
- Evaluated every model on held-out data across accuracy, precision, recall, F1 and MSE, with the best classifier reaching **0.75 F1 at 0.87 recall**.

EDUCATION

Arizona State University <i>MS, Software Engineering, Minor: Data Science, GPA: 3.7</i>	Aug 2024 – May 2026 <i>Tempe, AZ</i>
SRM Institute of Science and Technology <i>B.Tech, Computer Science</i>	Aug 2015 – May 2019 <i>Chennai, India</i>